

Story of How POSTECH Defeats KAIST

Mathematics for Science Quiz

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Preface

This book is devoted to the Mathematics section of Science Quiz in the **POSTECH**–**KAIST** Science War. It collects mathematical concepts, facts to memorize, past problems, and expected problems that are useful for competition preparation.

The main purpose of this text is not merely to record solutions, but to build a practical and reliable mathematical toolkit for Science Quiz. It includes standard theorems, important terminology, memorable facts about mathematicians and mathematical history, and problem-solving strategies that may be useful in an actual match.

I sincerely hope that this book contributes to **POSTECH**'s victory. Good luck, and let the game begin.

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Notation and Conventions

Throughout this book, we use standard mathematical notation whenever possible. The conventions below are arranged to keep the exposition readable, unified, and consistent with the chapter structure of Part I.

General Conventions

Scalars are written in ordinary italic letters, such as a, b, c, λ, t . Vectors are written in bold lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Matrices are written in bold uppercase letters, such as $\mathbf{A}, \mathbf{B}, \mathbf{P}$. Unless otherwise stated, vectors are regarded as column vectors.

Notation	Meaning
$\mathbb{N}$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{C}$	the set of complex numbers
$A \subseteq B$	A is a subset of B
$A \subsetneq B$	A is a proper subset of B
$A \setminus B$	the set difference of A and B
$ A $	the cardinality of a finite set A
$\emptyset$	the empty set
λ	usually a scalar, eigenvalue, or parameter, depending on context
ε	a small positive real number, especially in analysis
$\sim$	may mean asymptotic equivalence, distribution, or an equivalence relation
$O(\cdot)$	Big-O notation
$o(\cdot)$	little-o notation
$\square$	end of a proof or solution

Calculus, Analysis, and Topology

Calculus notation is used for derivatives, integrals, and elementary limits. Analysis and topology notation is used mainly for sequences, sets in metric spaces, compactness, connectedness, and continuous maps.

Notation	Meaning
$f'(a)$	the derivative of f at a

Notation	Meaning
$f^{(n)}(a)$	the n th derivative of f at a
$\frac{dy}{dx}$	the derivative of y with respect to x
$\int_a^b f(x) dx$	the definite integral of f from a to b
$C^k(I)$	the set of k times continuously differentiable functions on I
$C^\infty(I)$	the set of smooth functions on I
(a_n)	a sequence
$a_n \rightarrow a$	the sequence (a_n) converges to a
$\limsup_{n \rightarrow \infty} a_n$	the limit superior of (a_n)
$\liminf_{n \rightarrow \infty} a_n$	the limit inferior of (a_n)
$\bar{E}$	the closure of a set E
E°	the interior of a set E
∂E	the boundary of a set E
E'	the derived set, or the set of limit points of E
$B(p, r)$	the open ball centered at p with radius r
X, Y	usually topological spaces or metric spaces
U, V	usually open sets
K	often a compact set
$f : X \rightarrow Y$	a function from X to Y
$f^{-1}(E)$	the inverse image of E under f
$f(E)$	the image of E under f
$X \simeq Y$	X and Y are homeomorphic, when used in topology

We write intervals in the usual way:

$$(a, b), \quad [a, b], \quad [a, b), \quad (a, b].$$

In particular, the distinction between open and closed endpoints matters.

Linear Algebra

In linear algebra, we consistently use bold notation for vectors and matrices. For example, $\mathbf{v} \in \mathbb{R}^n$ is a vector and $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a matrix. The scalar λ is an eigenvalue, not a vector.

Notation	Meaning
$\mathbf{A}^\top$	the transpose of $\mathbf{A}$
$\mathbf{I}_n$	the $n \times n$ identity matrix
$\mathbf{0}$	the zero vector or zero matrix, depending on context
$\det(\mathbf{A})$	the determinant of $\mathbf{A}$
$\text{rk}(\mathbf{A})$	the rank of $\mathbf{A}$
$\text{tr}(\mathbf{A})$	the trace of $\mathbf{A}$
$\text{adj}(\mathbf{A})$	the adjugate of $\mathbf{A}$
$\text{Col}(\mathbf{A})$	the column space of $\mathbf{A}$
$\text{Null}(\mathbf{A})$	the null space of $\mathbf{A}$
$\text{Col}(\mathbf{A}^\top)$	the row space of $\mathbf{A}$
$\text{Row}(\mathbf{A})$	another notation for the row space, used only when convenient
$\text{Ker}(T)$	the kernel of a linear map T

Notation	Meaning
$\text{Im}(T)$	the image of a linear map T
$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	the span of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$
$\langle \mathbf{u}, \mathbf{v} \rangle$	the inner product of $\mathbf{u}$ and $\mathbf{v}$
$\ \mathbf{v}\ $	the norm of $\mathbf{v}$
$\mathbf{u} \perp \mathbf{v}$	$\mathbf{u}$ and $\mathbf{v}$ are orthogonal
$U \oplus V$	the direct sum of subspaces U and V

The four fundamental subspaces of $\mathbf{A} \in \mathbb{R}^{m \times n}$ are written as

$$\text{Col}(\mathbf{A}), \quad \text{Null}(\mathbf{A}), \quad \text{Col}(\mathbf{A}^\top), \quad \text{Null}(\mathbf{A}^\top).$$

We use the standard orthogonal decompositions

$$\mathbb{R}^m = \text{Col}(\mathbf{A}) \oplus \text{Null}(\mathbf{A}^\top), \quad \mathbb{R}^n = \text{Col}(\mathbf{A}^\top) \oplus \text{Null}(\mathbf{A}).$$

A projection matrix is denoted by $\mathbf{P}$ and satisfies

$$\mathbf{P}^2 = \mathbf{P}.$$

An orthogonal projection additionally satisfies

$$\mathbf{P}^\top = \mathbf{P}.$$

When a matrix is written as a product

$$\mathbf{A} = \mathbf{U}\mathbf{V},$$

the sizes of $\mathbf{U}$ and $\mathbf{V}$ should be inferred from context. Rank-one decompositions may be written as

$$\mathbf{A} = \sum_{i=1}^r \mathbf{u}_i \mathbf{v}_i^\top.$$

For cofactors, we write A_{ij} for the scalar cofactor of the entry a_{ij} of $\mathbf{A}$. The cofactor matrix is denoted by

$$\mathbf{C} = (A_{ij}),$$

and the adjugate matrix is

$$\text{adj}(\mathbf{A}) = \mathbf{C}^\top.$$

Thus

$$(\text{adj}(\mathbf{A}))_{ij} = A_{ji}.$$

We avoid using $\mathbf{A}^*$ for the adjugate, since $\mathbf{A}^*$ often denotes the conjugate transpose in other contexts.

Differential Equations

Differential equations are written using standard derivative notation. The independent variable is usually t or x , depending on the context.

Notation	Meaning
y'	the first derivative of y with respect to the relevant variable
y''	the second derivative of y with respect to the relevant variable
$\dot{x}$	the derivative of x with respect to time t
$\ddot{x}$	the second derivative of x with respect to time t
y_h	the homogeneous part of a solution
y_p	a particular solution
$c_1, c_2, \dots$	arbitrary constants in a general solution

Combinatorics, Discrete Mathematics, Probability and Statistics

Combinatorics and discrete mathematics provide the notation for counting, recurrence relations, and graphs. Probability and statistics use this notation frequently, especially in finite probability spaces.

Notation	Meaning
$n!$	factorial
$\binom{n}{k}$	the number of k -element subsets of an n -element set
$P(n, k)$	the number of k -permutations of an n -element set
a_n	the n th term of a sequence
$G = (V, E)$	a graph with vertex set V and edge set E
$\deg(v)$	the degree of a vertex v in a graph
K_n	the complete graph on n vertices
C_n	the cycle graph on n vertices
P_n	the path graph on n vertices
$\mathbb{P}(A)$	the probability of an event A
$\mathbb{P}(A \mid B)$	the conditional probability of A given B
$\mathbb{E}[X]$	the expectation of a random variable X
$\text{Var}(X)$	the variance of X
$\text{Cov}(X, Y)$	the covariance of X and Y
$X \sim \text{Bin}(n, p)$	X follows a binomial distribution with parameters n and p
$X \sim \text{Poi}(\lambda)$	X follows a Poisson distribution with parameter λ
$X \sim \text{N}(\mu, \sigma^2)$	X follows a normal distribution with mean μ and variance σ^2

Random variables are usually denoted by uppercase letters such as X, Y, Z . Events are denoted by uppercase letters such as A, B, E . In graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Geometry

We use standard geometric notation unless otherwise stated.

Notation	Meaning
$\triangle ABC$	the triangle with vertices A, B, C
$\angle ABC$	the angle with vertex B
$\overline{AB}$	the line segment joining A and B

Notation	Meaning
AB	the length of the segment $\overline{AB}$, when no confusion is possible
$[ABC]$	the area of triangle ABC , when used in geometry problems
$\parallel$	parallel
$\perp$	perpendicular

Number Theory and Algebra

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$. In algebra, the notation is kept standard and lightweight, since Science Quiz rarely requires highly specialized algebraic formalism.

Notation	Meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$\gcd(a, b)$	the greatest common divisor of a and b
$\text{lcm}(a, b)$	the least common multiple of a and b
$a \equiv b \pmod{n}$	a is congruent to b modulo n
$\varphi(n)$	Euler's phi function, or Euler's totient function
$\text{ord}_n(a)$	the order of a modulo n
p	usually a prime number, when used in number-theoretic contexts
G	usually a group
e	the identity element of a group, when the context is clear
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
G/H	the quotient group of G by H
R	usually a ring
F or K	usually a field
$F[x]$	the polynomial ring over a field F
$\deg f$	the degree of a polynomial f
$\text{Aut}(G)$	the automorphism group of G
$\text{Gal}(L/K)$	the Galois group of a field extension L/K

The Chinese Remainder Theorem may be abbreviated as CRT after it is first introduced. In algebraic contexts, $\deg f$ denotes the degree of a polynomial f ; in graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Miscellaneous Topics

Some symbols have different meanings in different areas. In this book, the intended meaning should always be clear from context.

Notation	Meaning
$\lfloor x \rfloor$	the floor of x
$\lceil x \rceil$	the ceiling of x
$\{x\}$	the fractional part of x , usually $x - \lfloor x \rfloor$

Part I

Mathematical Concepts

Chapter 1

Calculus

This chapter collects the calculus results that are most likely to be useful in Science Quiz: standard limits, continuity theorems, differentiation principles, integration tools, Taylor expansions, and a few famous special formulas.

Formula 1.1. Standard Limits

The following limits are fundamental:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1, & \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \frac{1}{2}, \\ \lim_{x \rightarrow 0} \frac{e^x - 1}{x} &= 1, & \lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} &= 1,\end{aligned}$$

and

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

Strategy. These limits are often faster than L'Hopital's Rule. In a quiz, the expressions $\sin x$, $1 - \cos x$, $e^x - 1$, $\log(1 + x)$, and $(1 + x)^{1/x}$ are strong signals for these standard patterns.

Theorem 1.2. Squeeze Theorem

Suppose that

$$f(x) \leq g(x) \leq h(x)$$

for all x sufficiently close to a , except possibly at a itself. If

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Definition 1.3. Continuity

A function f is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

We say that f is continuous on an interval if it is continuous at every point of that interval.

Theorem 1.4. Intermediate Value Theorem

Let f be continuous on $[a, b]$. If N is any number between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that

$$f(c) = N.$$

Strategy. The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values have opposite signs at two endpoints.

Theorem 1.5. Extreme Value Theorem

Let f be continuous on $[a, b]$. Then f attains both a maximum and a minimum on $[a, b]$. That is, there exist $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d)$$

for all $x \in [a, b]$.

Remark. The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

Definition 1.6. Derivative

Let f be a real-valued function. The derivative of f at a is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided that this limit exists.

Proposition 1.7. Basic Differentiation Rules

If f and g are differentiable, then

$$(f+g)' = f' + g', \quad (fg)' = f'g + fg',$$

and, where $g \neq 0$,

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

If g is differentiable at x and f is differentiable at $g(x)$, then

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

Theorem 1.8. Fermat's Theorem on Stationary Points

Suppose that f has a local maximum or local minimum at an interior point c of its domain. If f is differentiable at c , then

$$f'(c) = 0.$$

Theorem 1.9. Rolle's Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) . If

$$f(a) = f(b),$$

then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

Theorem 1.10. Mean Value Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Idea. The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

Corollary 1.11. Monotonicity Test

Let f be differentiable on an interval I . If

$$f'(x) > 0$$

for all $x \in I$, then f is strictly increasing on I . If

$$f'(x) < 0$$

for all $x \in I$, then f is strictly decreasing on I .

Strategy. For one-variable optimization problems, check endpoints and then check interior critical points. Interior extrema usually come from solving $f'(x) = 0$.

Theorem 1.12. L'Hopital's Rule

Suppose that f and g are differentiable near a , and that

$$\lim_{x \rightarrow a} f(x) = 0, \quad \lim_{x \rightarrow a} g(x) = 0.$$

If $g'(x) \neq 0$ near a and

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

The same principle also applies to many ∞/∞ indeterminate forms.

Theorem 1.13. Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$ and define

$$F(x) = \int_a^x f(t) dt.$$

Then F is differentiable on (a, b) and

$$F'(x) = f(x).$$

Also, if G is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = G(b) - G(a).$$

Formula 1.14. Substitution Rule

If $u = g(x)$, then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

In definite-integral form,

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Formula 1.15. Integration by Parts

If u and v are differentiable functions, then

$$\int u dv = uv - \int v du.$$

Equivalently,

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

Formula 1.16. Gaussian Integral

The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Equivalently,

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Remark. The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

Theorem 1.17. Taylor's Theorem

Suppose that f is $n + 1$ times differentiable near a . Then

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x),$$

where, in Lagrange's form of the remainder,

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$$

for some c between a and x .

Strategy. Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving e^x , $\sin x$, $\cos x$, $\log(1 + x)$, or $\arctan x$ often invite Taylor expansion.

Formula 1.18. Maclaurin Series

The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1),$$

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \quad (-1 < x \leq 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad (|x| \leq 1).$$

Formula 1.19. Binomial Series

For any real number α ,

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \quad (|x| < 1),$$

where

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}.$$

Formula 1.20. Euler's Formula

For every real number x ,

$$e^{ix} = \cos x + i \sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

Remark. The identity $e^{i\pi} + 1 = 0$ is called Euler's identity. It connects the constants e , i , π , 1 , and 0 in a single formula.

Formula 1.21. Wallis Product

The Wallis product is

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots.$$

Definition 1.22. Gamma Function

For $x > 0$, the Gamma function is defined by

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt.$$

Proposition 1.23. Gamma Function and Factorials

The Gamma function satisfies

$$\Gamma(x+1) = x\Gamma(x).$$

In particular, for every positive integer n ,

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Formula 1.24. Stirling's Formula

As $n \rightarrow \infty$,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

Remark. Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

Chapter 2

Linear Algebra

Definition 2.1. Vector Space

A vector space over a field F is a set V equipped with vector addition and scalar multiplication satisfying the usual vector space axioms. Elements of V are called vectors, and elements of F are called scalars.

Remark. In this book, the most common examples are $\mathbb{R}^n$, $\mathbb{C}^n$, spaces of polynomials, and spaces of functions.

Definition 2.2. Linear Independence, Span, and Basis

Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be vectors in a vector space V . They are linearly independent if

$$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0}$$

implies

$$c_1 = \dots = c_k = 0.$$

Their span is

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \{c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k : c_1, \dots, c_k \in F\}.$$

A basis of V is a linearly independent spanning set of V .

Definition 2.3. Dimension

If a vector space V has a finite basis, then all bases of V have the same number of vectors. This number is called the dimension of V and is denoted by $\dim V$.

Theorem 2.4. Rank–Nullity Theorem

Let $T : V \rightarrow W$ be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \text{Ker}(T) + \dim \text{Im}(T).$$

For a matrix $\mathbf{A} \in F^{m \times n}$, this becomes

$$n = \text{rk}(\mathbf{A}) + \dim \text{Null}(\mathbf{A}).$$

Strategy. Rank–Nullity is the fastest way to connect the number of variables, the number of pivots, and the number of free variables.

Theorem 2.5. Invertible Matrix Theorem

For a square matrix $\mathbf{A} \in F^{n \times n}$, the following statements are equivalent:

$\mathbf{A}$ is invertible,

$$\det(\mathbf{A}) \neq 0,$$

$$\text{rk}(\mathbf{A}) = n,$$

$$\text{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

has a unique solution for every $\mathbf{b} \in F^n$.

Remark. This theorem is not one single trick, but a dictionary. It lets us translate between determinants, rank, null spaces, and solvability of linear systems.

Definition 2.6. Determinant

The determinant is a scalar associated to a square matrix $\mathbf{A} \in F^{n \times n}$, denoted by $\det(\mathbf{A})$. Geometrically, for real matrices, $|\det(\mathbf{A})|$ is the volume scaling factor of the linear transformation $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$.

Proposition 2.7. Basic Properties of Determinants

For square matrices $\mathbf{A}, \mathbf{B} \in F^{n \times n}$, we have

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}),$$

$$\det(\mathbf{A}^\top) = \det(\mathbf{A}),$$

and, if $\mathbf{A}$ is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Formula 2.8. Determinant of a Triangular Matrix

If $\mathbf{A}$ is upper triangular or lower triangular with diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$, then

$$\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}.$$

Definition 2.9. Trace

The trace of a square matrix $\mathbf{A} = (a_{ij}) \in F^{n \times n}$ is

$$\text{tr}(\mathbf{A}) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Proposition 2.10. Trace Identities

If $\mathbf{A}$ and $\mathbf{B}$ are matrices of compatible sizes, then

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}).$$

In particular, similar matrices have the same trace.

Definition 2.11. Eigenvalue and Eigenvector

Let $\mathbf{A} \in F^{n \times n}$. A scalar $\lambda \in F$ is an eigenvalue of $\mathbf{A}$ if there exists a nonzero vector $\mathbf{v} \in F^n$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

Such a nonzero vector $\mathbf{v}$ is called an eigenvector corresponding to λ .

Definition 2.12. Characteristic Polynomial

For a square matrix $\mathbf{A} \in F^{n \times n}$, the polynomial

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A})$$

is called the characteristic polynomial of $\mathbf{A}$.

Proposition 2.13. Eigenvalues and the Characteristic Polynomial

A scalar λ is an eigenvalue of $\mathbf{A}$ if and only if

$$\det(\lambda\mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial $p_{\mathbf{A}}(\lambda)$.

Strategy. When a quiz problem asks for eigenvalues, first compute the characteristic polynomial. When it asks for eigenvectors, solve

$$(\mathbf{A} - \lambda \mathbf{I}_n)\mathbf{x} = \mathbf{0}.$$

Proposition 2.14. Trace, Determinant, and Eigenvalues

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ have eigenvalues $\lambda_1, \dots, \lambda_n$, counted with algebraic multiplicity. Then

$$\text{tr}(\mathbf{A}) = \lambda_1 + \dots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \dots \lambda_n.$$

Theorem 2.15. Cayley–Hamilton Theorem

Every square matrix satisfies its own characteristic polynomial. That is, if

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}),$$

then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

Remark. Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

Definition 2.16. Diagonalizable Matrix

A matrix $\mathbf{A} \in F^{n \times n}$ is diagonalizable over F if there exist an invertible matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}.$$

Theorem 2.17. Diagonalization Criterion

A matrix $\mathbf{A} \in F^{n \times n}$ is diagonalizable over F if and only if F^n has a basis consisting of eigenvectors of $\mathbf{A}$.

Corollary 2.18. Distinct Eigenvalues Imply Diagonalizability

If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues in F , then $\mathbf{A}$ is diagonalizable over F .

Strategy. Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

Theorem 2.19. Spectral Theorem for Real Symmetric Matrices

If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, then all eigenvalues of $\mathbf{A}$ are real, and there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^T.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Remark. The Spectral Theorem is one of the most important theorems in linear algebra. In quiz settings, the words “real symmetric matrix” are a strong signal for real eigenvalues and orthogonal diagonalization.

Definition 2.20. Inner Product and Norm

In $\mathbb{R}^n$, the standard inner product of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

The associated norm is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Theorem 2.21. Cauchy–Schwarz Inequality

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Equivalently, for real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Strategy. In a quiz setting, the expression $\sum a_i b_i$ is a strong signal for Cauchy–Schwarz. The vector form is often the fastest way to recognize the inequality.

Corollary 2.22. Triangle Inequality

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Definition 2.23. Orthogonal Matrix

A real square matrix $\mathbf{Q}$ is orthogonal if

$$\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}.$$

Equivalently,

$$\mathbf{Q}^{-1} = \mathbf{Q}^\top.$$

Proposition 2.24. Properties of Orthogonal Matrices

If $\mathbf{Q}$ is orthogonal, then it preserves inner products and norms:

$$\langle \mathbf{Q}\mathbf{u}, \mathbf{Q}\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle,$$

and

$$\|\mathbf{Q}\mathbf{v}\| = \|\mathbf{v}\|.$$

Also,

$$\det(\mathbf{Q}) = \pm 1.$$

Theorem 2.25. Gram–Schmidt Process

From any linearly independent list

$$\mathbf{v}_1, \dots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \dots, \mathbf{q}_k$$

with the same span:

$$\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

Remark. Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

Definition 2.26. Orthogonal Projection

Let W be a subspace of $\mathbb{R}^n$. The orthogonal projection of $\mathbf{v} \in \mathbb{R}^n$ onto W is the unique vector $\text{proj}_W(\mathbf{v}) \in W$ such that

$$\mathbf{v} - \text{proj}_W(\mathbf{v}) \perp W.$$

Formula 2.27. Projection onto a Line

If $\mathbf{u} \neq \mathbf{0}$, then the projection of $\mathbf{v}$ onto the line spanned by $\mathbf{u}$ is

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}.$$

Formula 2.28. Projection Matrix

If the columns of $\mathbf{Q} \in \mathbb{R}^{n \times k}$ are orthonormal, then the orthogonal projection matrix onto $\text{Col}(\mathbf{Q})$ is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top.$$

More generally, if the columns of $\mathbf{A} \in \mathbb{R}^{n \times k}$ are linearly independent, then the orthogonal projection matrix onto $\text{Col}(\mathbf{A})$ is

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top.$$

Proposition 2.29. Projection Matrix Criterion

A matrix $\mathbf{P}$ is a projection matrix if

$$\mathbf{P}^2 = \mathbf{P}.$$

It is an orthogonal projection matrix if, in addition,

$$\mathbf{P}^\top = \mathbf{P}.$$

Definition 2.30. Singular Values

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^\top \mathbf{A}.$$

Theorem 2.31. Singular Value Decomposition

Every real matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be written as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top,$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices and $\mathbf{\Sigma}$ is a rectangular diagonal matrix whose diagonal entries are the singular values of $\mathbf{A}$.

Remark. Singular Value Decomposition, or SVD, is one of the most powerful decompositions in linear algebra. It applies to every real matrix, not only to square or diagonalizable matrices.

Theorem 2.32. Jordan Canonical Form

Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every $\mathbf{A} \in \mathbb{C}^{n \times n}$, there exists an invertible matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$$

is in Jordan canonical form.

Remark. Jordan canonical form is usually more important as a named concept than as a computational tool in Science Quiz. It describes what remains when a matrix cannot be fully diagonalized.

Chapter 3

Differential Equations

Definition 3.1. Differential Equation

A differential equation is an equation involving an unknown function and one or more of its derivatives. An ordinary differential equation, or ODE, involves derivatives with respect to one independent variable. A partial differential equation, or PDE, involves partial derivatives with respect to several independent variables.

Definition 3.2. Order of a Differential Equation

The order of a differential equation is the highest order of derivative appearing in the equation. For example,

$$y'' + 3y' - 2y = 0$$

is a second-order differential equation.

Definition 3.3. Linear Differential Equation

A differential equation is linear if the unknown function and its derivatives appear only linearly. For example,

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

is a linear differential equation.

Remark. Linearity is a major dividing line. Linear equations have a powerful superposition principle, while nonlinear equations may behave much more unpredictably.

Theorem 3.4. Existence and Uniqueness Theorem

Consider the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0.$$

If f and $\partial f / \partial y$ are continuous near (t_0, y_0) , then there exists a unique solution near $t = t_0$.

Remark. In a quiz setting, the important phrase is “existence and uniqueness.” It tells us when an initial condition determines exactly one solution.

Formula 3.5. Separable Equation

A first-order differential equation of the form

$$\frac{dy}{dt} = g(t)h(y)$$

is called separable. It can be solved by writing

$$\frac{1}{h(y)} dy = g(t) dt$$

and integrating both sides.

Strategy. If the variables can be separated so that all y -terms are on one side and all t -terms are on the other, try separation of variables first.

Formula 3.6. First-Order Linear Equation

A first-order linear equation has the form

$$y' + p(t)y = q(t).$$

Its integrating factor is

$$\mu(t) = e^{\int p(t) dt}.$$

Multiplying by $\mu(t)$ gives

$$(\mu(t)y(t))' = \mu(t)q(t),$$

so

$$y(t) = \frac{1}{\mu(t)} \left(\int \mu(t)q(t) dt + C \right).$$

Definition 3.7. Homogeneous Linear Equation

A linear differential equation is homogeneous if its forcing term is zero. For example,

$$a_n(t)y^{(n)} + \cdots + a_1(t)y' + a_0(t)y = 0$$

is homogeneous. If the right-hand side is not zero, the equation is nonhomogeneous.

Theorem 3.8. Superposition Principle

If $y_1, \dots, y_k$ are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1y_1 + \cdots + c_ky_k$$

is also a solution.

Remark. The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

Proposition 3.9. General Solution of a Nonhomogeneous Linear Equation

For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is one particular solution of the nonhomogeneous equation.

Strategy. For linear nonhomogeneous equations, remember the phrase

$$\text{general solution} = \text{homogeneous solution} + \text{particular solution}.$$

Formula 3.10. Second-Order Constant-Coefficient Equation

Consider the homogeneous equation

$$ay'' + by' + cy = 0, \quad a \neq 0.$$

Its characteristic equation is

$$ar^2 + br + c = 0.$$

If the roots are distinct real numbers r_1, r_2 , then

$$y_h = c_1e^{r_1t} + c_2e^{r_2t}.$$

If the equation has a repeated real root r , then

$$y_h = (c_1 + c_2t)e^{rt}.$$

If the roots are $\alpha \pm i\beta$ with $\beta \neq 0$, then

$$y_h = e^{\alpha t} (c_1 \cos \beta t + c_2 \sin \beta t).$$

Remark. The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

Definition 3.11. Wronskian

For two differentiable functions y_1 and y_2 , their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Proposition 3.12. Wronskian and Linear Independence

If

$$W(y_1, y_2)(t_0) \neq 0$$

for some t_0 , then y_1 and y_2 are linearly independent.

Definition 3.13. Equilibrium Solution

For an autonomous equation

$$y' = f(y),$$

an equilibrium solution is a constant solution $y(t) = y_0$ satisfying

$$f(y_0) = 0.$$

Strategy. For autonomous equations, equilibrium solutions come from setting the derivative equal to zero. Then the sign of $f(y)$ tells us whether nearby solutions move upward or downward.

Formula 3.14. Exponential Growth and Decay

The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If $k > 0$, this models exponential growth. If $k < 0$, this models exponential decay.

Formula 3.15. Logistic Equation

The logistic equation is

$$y' = ky \left(1 - \frac{y}{K}\right),$$

where $k > 0$ is the growth rate and $K > 0$ is the carrying capacity. Its equilibrium solutions are

$$y = 0 \quad \text{and} \quad y = K.$$

Remark. The logistic equation is the standard model for population growth with limited resources. The parameter K represents the limiting population.

Formula 3.16. Simple Harmonic Oscillator

The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A \cos(\omega t - \phi)$$

for suitable constants A and ϕ .

Remark. This is the equation behind ideal mass-spring motion and many small oscillation models in physics.

Formula 3.17. Damped Harmonic Oscillator

The equation

$$mx'' + cx' + kx = 0$$

models a damped harmonic oscillator. Its characteristic equation is

$$mr^2 + cr + k = 0.$$

The motion is underdamped, critically damped, or overdamped according as

$$c^2 - 4mk < 0, \quad c^2 - 4mk = 0, \quad c^2 - 4mk > 0.$$

Definition 3.18. Laplace Transform

The Laplace transform of a function $f(t)$ is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t) dt.$$

Proposition 3.19. Basic Laplace Transforms

The most common Laplace transforms are

$$\mathcal{L}\{1\}(s) = \frac{1}{s}, \quad \mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a},$$

$$\mathcal{L}\{\sin at\}(s) = \frac{a}{s^2 + a^2}, \quad \mathcal{L}\{\cos at\}(s) = \frac{s}{s^2 + a^2}.$$

Remark. The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

Definition 3.20. Heat Equation

The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \quad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

Definition 3.21. Wave Equation

The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed c .

Theorem 3.22. d'Alembert's Formula

The solution of the wave equation

$$u_{tt} = c^2 u_{xx}$$

on the real line with initial conditions

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x),$$

is

$$u(x, t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds.$$

Remark. d'Alembert's formula shows that waves move along the characteristic lines $x - ct = \text{constant}$ and $x + ct = \text{constant}$.

Definition 3.23. Laplace Equation

Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

Definition 3.24. Poisson Equation

Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

Remark. The heat equation, wave equation, and Laplace equation are three of the most famous PDEs. A quiz problem may ask for their names, forms, or physical interpretations.

Chapter 4

Combinatorics and Discrete Mathematics

Chapter 5

Probability and Statistics

Chapter 6

Geometry

Chapter 7

Number Theory

Chapter 8

Algebra

Chapter 9

Analysis

Chapter 10

Topology

Chapter 11

Miscellaneous Topics

Formula 11.1. Mercator Series for $\ln 2$

The alternating harmonic series satisfies

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2.$$

Remark. This follows from the Mercator series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

by substituting $x = 1$.

Formula 11.2. Basel Sum

The Basel sum is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}.$$

Equivalently,

$$\zeta(2) = \frac{\pi^2}{6}.$$

Remark. The problem of evaluating

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is called the Basel problem. Its famous solution was found by Euler.

Formula 11.3. Odd Part of the Basel Sum

The sum of reciprocal squares over odd positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

Idea. This is obtained by subtracting the even part from the Basel sum:

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \left(1 - \frac{1}{4}\right) \frac{\pi^2}{6} = \frac{\pi^2}{8}.$$

Formula 11.4. Alternating Basel Series

The alternating Basel series satisfies

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}.$$

Equivalently,

$$\eta(2) = \frac{\pi^2}{12},$$

where $\eta(s)$ is the Dirichlet eta function.

Idea. The positive terms are the odd reciprocal squares and the negative terms are the even reciprocal squares. Hence

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

Formula 11.5. Gregory–Leibniz Series

The Gregory–Leibniz series is

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2n-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

Remark. This follows from the Taylor series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

by substituting $x = 1$, since $\arctan 1 = \pi/4$.

Strategy. These values are useful because Science Quiz problems often hide them behind a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right) = \frac{\ln 2}{2}.$$

Part II

Facts to Memorize

Chapter 12

ICM and the Fields Medal

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.

Year	ICM location	Medalists and areas	Notes
1936	Oslo, Norway	Lars Ahlfors — complex analysis, Riemann surfaces; Jesse Douglas — minimal surfaces, Plateau problem	First Fields Medals.
1950	Cambridge, USA	Laurent Schwartz — distribution theory; Atle Selberg — analytic number theory, sieve methods, zeta functions	First awards after World War II.
1954	Amsterdam, Netherlands	Kunihiko Kodaira — complex geometry, algebraic geometry; Jean-Pierre Serre — algebraic topology, sheaf theory	Serre is the youngest Fields Medalist.
1958	Edinburgh, UK	Klaus Roth — number theory, Diophantine approximation; Rene Thom — cobordism, differential topology	
1962	Stockholm, Sweden	Lars Hormander — partial differential equations; John Milnor — differential topology, exotic spheres	
1966	Moscow, USSR	Michael Atiyah — K-theory, index theorem; Paul Cohen — set theory, forcing, continuum hypothesis; Alexander Grothendieck — algebraic geometry, homological algebra; Stephen Smale — differential topology, generalized Poincare conjecture	First year with four medalists.
1970	Nice, France	Alan Baker — transcendental number theory; Heisuke Hironaka — algebraic geometry, resolution of singularities; Serge Novikov — topology, differential geometry; John G. Thompson — finite group theory	
1974	Vancouver, Canada	Enrico Bombieri — number theory, complex analysis, minimal surfaces; David Mumford — algebraic geometry, moduli spaces	
1978	Helsinki, Finland	Pierre Deligne — algebraic geometry, Weil conjectures; Charles Fefferman — analysis, several complex variables; Grigory Margulis — Lie groups, ergodic theory, discrete groups; Daniel Quillen — algebraic K-theory	

Continued on the next page.

Year	ICM location	Medalists and areas	Notes
1982	Warsaw, Poland	Alain Connes — operator algebras, noncommutative geometry; William Thurston — low-dimensional topology, 3-manifolds; Shing-Tung Yau — differential geometry, PDE, Calabi conjecture	ICM Warsaw was held in 1983.
1986	Berkeley, USA	Simon Donaldson — four-manifold topology, gauge theory; Gerd Faltings — arithmetic algebraic geometry, Mordell conjecture; Michael Freedman — four-manifold topology, four-dimensional Poincare conjecture	
1990	Kyoto, Japan	Vladimir Drinfeld — quantum groups, Langlands program; Vaughan Jones — operator algebras, knot theory; Shigefumi Mori — algebraic geometry, minimal model program; Edward Witten — mathematical physics, topology, quantum field theory	Witten was the first physicist to receive the Fields Medal.
1994	Zurich, Switzerland	Jean Bourgain — analysis, Banach spaces, harmonic analysis; Pierre-Louis Lions — partial differential equations; Jean-Christophe Yoccoz — dynamical systems; Efim Zelmanov — algebra, restricted Burnside problem	
1998	Berlin, Germany	Richard Borcherds — algebra, automorphic forms, moonshine; W. Timothy Gowers — functional analysis, combinatorics; Maxim Kontsevich — algebraic geometry, topology, mathematical physics; Curtis T. McMullen — complex dynamics, hyperbolic geometry	Andrew Wiles received a special IMU silver plaque.
2002	Beijing, China	Laurent Lafforgue — Langlands correspondence over function fields; Vladimir Voevodsky — motivic cohomology, $\mathbb{A}^1$ -homotopy theory	
2006	Madrid, Spain	Andrei Okounkov — probability, representation theory, algebraic geometry; Grigori Perelman — Ricci flow, Poincare conjecture; Terence Tao — PDE, harmonic analysis, additive combinatorics; Wendelin Werner — probability, SLE, Brownian motion	Perelman declined the medal.
2010	Hyderabad, India	Elon Lindenstrauss — ergodic theory, number theory; Ngo Bao Chau — automorphic forms, Fundamental Lemma; Stanislav Smirnov — probability, statistical physics, conformal invariance; Cedric Villani — kinetic theory, Boltzmann equation, optimal transport	
2014	Seoul, South Korea	Artur Avila — dynamical systems; Manjul Bhargava — number theory, geometry of numbers; Martin Hairer — stochastic partial differential equations; Maryam Mirzakhani — Riemann surfaces, moduli spaces, dynamics	Mirzakhani was the first woman Fields Medalist.
2018	Rio de Janeiro, Brazil	Caucher Birkar — algebraic geometry, Fano varieties, minimal model program; Alessio Figalli — optimal transport, PDE, metric geometry; Peter Scholze — arithmetic algebraic geometry, p -adic geometry; Akshay Venkatesh — number theory, homogeneous dynamics, representation theory	

Continued on the next page.

Year	ICM location	Medalists and areas	Notes
2022	Helsinki, Finland	Hugo Duminil-Copin — probability, statistical physics, phase transitions; June Huh — Hodge theory, combinatorics, matroids; James Maynard — analytic number theory, prime numbers; Maryna Viazovska — sphere packing, E_8 lattice, Fourier analysis	ICM 2022 was virtual; the award ceremony was held in Helsinki. Viazovska was the second woman Fields Medalist.

Essential ICM and Fields Medal Landmarks

- ICM stands for International Congress of Mathematicians.
- IMU stands for International Mathematical Union.
- The Fields Medal is awarded every four years at the ICM.
- The Fields Medal is named after John Charles Fields.
- The Fields Medal was first awarded in 1936.
- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal.
- Andrew Wiles received a special IMU silver plaque in 1998 for his proof of Fermat's Last Theorem.
- Grigori Perelman declined the Fields Medal in 2006 after proving the Poincare conjecture.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- The 2026 ICM is scheduled to be held in Philadelphia, USA.

Part III

Past Problems

Problem 2018.1. The following are this year's Fields Medalists. Excluding choice (5), who is the second oldest among them?

- (1) Caucher Birkar
- (2) Alessio Figalli
- (3) Peter Scholze
- (4) Akshay Venkatesh
- (5) The person who stole Birkar's medal

Solution. The 2018 Fields Medalists were

- (1) Caucher Birkar born in 1978,
- (2) Alessio Figalli born in 1984,
- (3) Peter Scholze born in 1987,
- (4) Akshay Venkatesh born in 1981.

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

(4).

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal. □

Problem 2018.2. According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of n possible geometric structures. What is n ?

Solution. The 3-dimensional geometric structures appearing in Thurston's geometrization conjecture are the eight Thurston geometries:

$$S^3, \quad \mathbb{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}_2(\mathbb{R})}, \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence $n = 8$, so the answer is

8.

□

Part IV

Expected Problems